Second-round review: *Composition is a nerve-level-two datum* and *A uniqueness theorem for the container derivative* (MacBeth 6089268)

Rick

To: MacBeth (Kodamai)

2026-10-11

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Date:	2026-10-11
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Title:	Round-2 review: nerve-level-two note (A), uniqueness note (B)
Reviewed commit:	content `6089268` (stamp `6330238`), `scotmacbeth/work-in-progress`; both PDFs print "6089268" on p. 1. A: 13 pp. B: 17 pp.
Previous reports:	A: 2026-10-09 on `242c7ea`. B: 2026-10-09 on `75603ad`.
Our WIP commit:	`fc69dab` (grandpa-rick/work-in-progress)
Scripts:	`reviews/scripts/2026-10-11-nerve-r2/` (3 scripts + logs; all PASS)

"l." is the line in `pdftotext -layout` output of the 6089268 PDFs. Grades: **computed** means my own code, written from your printed definitions, checked it. **checked-sober** means I read the argument by hand and it holds. **read** means I only read it. Verdicts on items: **MUST-FIX**, **SHOULD**, **OK**. I checked on GitHub that `6089268` exists and that `6330238` is its child ("stamp content commit 6089268").

Verdict

- **Note A (nerve level two): not yet.** N1 is fixed exactly. The N2 rewrite gets the direction right: a class presupposes the cocycle condition, and the class measures splitting. But the splitting derivation still has three gaps (A1). N2′ (A2) and the N3 example (A3) are unchanged. Three **MUST-FIX**, each a sentence or a paragraph. No new mathematics is needed.
- **Note B (uniqueness): accept after one edit.** I confirmed that the changes are wording only. No theorem, lemma or definition statement changed, except Proposition 12, which we asked you to change. But the §1 paragraph still says "$\mathbb{N} \cdot y$ is solid ... hence an isomorphism ℓ_M" (B1), and the §1 "exotic" sentence is still unscoped (B2). Both are 30-second edits.

Weld nodes. Do *not* re-promote them on this version. They can go to **checked-sober**, not *proved*, once A1 is in the text. The standard content of the gloss is right: the cocycle condition is associativity of the twisted composition, and for cocycles $[\omega] = 0$ iff the extension splits. I **computed** both directions exhaustively in four cases (§3). What is still missing is the bridge from that standard content to *your* weld: which hypothesis supplies associativity, which category is extended, and the condition (L).

1 Disposition of earlier points

Note A (our 2026-10-09 report on 242c7ea)

Item	Status	Evidence (6089268)
N1 $U \subsetneq \mathrm{tr}_1 N$	**FIXED**	p. 4 l.163–169: "strictly coarser shadow of $\mathrm{tr}_1 N(C)$, not an equal". Witness [1] vs $B(\mathbb{Z}/2) \sqcup \mathrm{pt}$. "equivalently" and "$= U(p)$" are gone (diff). Thm 3 l.280–281 and §2 l.213–214 are reworded. Witness **computed** (§3).
N2 cocycle $\neq$ class	**PARTIAL**	Direction fixed: abstract l.27–30, §2 l.247–254, Rem. 13 l.438–448 ("a class presupposes associativity"). The old "vanishing iff the weld is associative" and "witnessed by $[\omega] = \varepsilon$" are deleted from Thm 12. New gaps in the splitting derivation: A1.
N2′ what $[\omega]$ is a class *of*	**NOT FIXED**	Thm 14 l.467 still assumes $S = C \bowtie D$. Ex. 16 l.482–484 still uses $B(\mathbb{Z}/2)$ "whose weld class generates $H^2(S; \mathbb{Z}/2)$". No extended category and no C, D are named. With the new gloss the conflict is sharper: A2.
N3 two senses of "level"	**PARTIAL**	The *cochain level* is now defined (l.177–179) and used correctly in Rem. 2 (weld: cochain level 2, nerve level 3). But Rem. 8 (l.359–365) is byte-identical to 242c7ea, and it was our worked counter-example: A3.
Recipient on p. 1	**FIXED**	l.10 "Prepared for Rick (referee) and Neil Ghani (PI)".
W1 Rem. 8 "not even stably wrong"	**NOT FIXED**	l.364–365 unchanged.
W2 δ-translations vs $H^1(-; \mathbb{F}_2)$	**NOT FIXED**	Abstract l.24–26 and §7 l.514–516 unchanged. l.573–575 still says "underwritten by Lemma 1 and Theorems 5, 7 and 9" (our old 1.3 residue).
W3 Thm 11 is classical	**PARTIAL**	Title now says "2-coskeletal collapse", and l.425 mentions Segal. Still credited to [7] and counted as "a collapse theorem" that is proved (l.122, l.530).
W4 Lemma 1 is a definition	**NOT FIXED**	Still a "regrade lemma" among the proved results. Nice-to-have only.
W5 Thm 9 base-point phrasing	**NOT FIXED**	l.376–377 unchanged.
Diamond C, D in the example	**NOT FIXED**	Ex. 16 l.478–481 unchanged.

Note B (our 2026-10-09 report on 75603ad)

Item	Status	Evidence (6089268)
M1 unscoped "only nontrivial"	**PARTIAL**	§8 (p. 15 l.766–767, "among representable Weil-base-change tangent structures") and §9 (p. 16 l.803–805) are fixed. §1 l.75–76 is unchanged: "Could there be an exotic first-order tangent bundle . . . ? This note proves there cannot." See B2.
M2 vacuous footnote 1	**FIXED**	p. 10 footnote: "no bijection $K \cong K \times K$ at all . . . nothing to search over". It correctly assigns the content to steps (4)–(5) at infinite rank. **Computed**.
M3 "$\mathbb{N} \cdot y$ is solid"	**PARTIAL**	Prop. 12, §5 l.426–431, Rem. 15 and §9 are fixed. But §1 p. 4 l.177–178 still reads "It is true that $\mathbb{N} \cdot y$ is solid: the Cantor pairing . . . supplies an isomorphism $M \otimes M \cong M$, *hence an isomorphism* ℓ_M". Also l.167 "solid but not a structure", l.183 and l.202 "Solidity is . . . strictly weaker". See B1.
M4 recipient	**FIXED**	p. 1 "Prepared for Rick (referee) and Neil Ghani (PI)". The date line now ends "Revised 10 October".
S1 quote CC14 axiom; credit universality	**PARTIAL**	Step (4) l.497–503 adds "the c-twist term is idle". The axiom is still written $T(\ell) \circ \ell = (c\text{-twist}) \circ \ell_T \circ \ell$, not quoted. Step (3) is not demoted. "the flip and lift-coassociativity axioms" is still the credited mechanism at l.57, l.166, l.180 and l.789–790.
S2 derive v; $\delta(1) = 1 \otimes 1$	**NOT FIXED**	Step (1) l.469–474 unchanged.
S3 Thm 18 closing; Rem. 23	**NOT FIXED**	The new cite "the same input as step (1) of Theorem 13" is attached to the *opening* universality sentence (l.633). The closing l.645–650 still says "as one exhibits directly . . . $\ell_M : \varepsilon \mapsto \varepsilon \otimes \varepsilon$ (the $\varepsilon \mapsto \varepsilon_1 \varepsilon_2$ of $W \otimes W$)". Rem. 23 l.719–720 still says "pinned by $\varepsilon \mapsto \varepsilon_1 \varepsilon_2$ and $c\ell = \ell$". **Computed**: $c\ell = \ell$ and coassociativity hold for every k in $\delta(\varepsilon) = k\,\varepsilon \otimes \varepsilon$. Only universality forces $k = 1$.
S4 novelty tone-down	**NOT FIXED**	l.42 "not only faithful but cleaner"; l.137 "the one new idea"; l.385 "strictly outside the Lanfranchi–Lemay hypotheses"; l.732, l.764 "isolating exactly which uses of ring structure . . . were essential (none . . .)".
S5 Lemma 7/9	**NOT FIXED**	Lemma 9 proof l.376–379 ("morphisms between linear containers are just functions on shapes") unchanged.
C1–C7, nits	**NOT FIXED**	C1 l.225; C2 l.80 vs l.237; C3 l.653; C4 l.665; C6 l.124; C7 Rem. 23. "irreducible" still lacks $x \neq 0$ (l.481). $n(n-1)$ is still at l.400. Rem. 3 "unit of the adjunction" is still at l.269. Step (3)'s "no exotic non-representable flip" is still at l.491–492.

Your covering email says, for B, "w1–w4, wording only". That is accurate for what was changed. It does not cover S2, S3, S4 and S5, which your 2026-10-10 acknowledgement accepted ("Thm 18's closing will cite Thm 13 step (1)"). Only the opening sentence of Theorem 18 got the citation.

2 The splitting derivation, line by line (Note A)

I reconstruct the derivation as printed, in the order a reader meets it, and grade each step. The model behind every step is the twisted composition on $S \times D$,

$$(x, u) \cdot (y, v) = (xy,\ u + x{\cdot}v + \omega(x, y)),$$

over a base category S with coefficient system D. This is the Eilenberg–Mac Lane extension or, for categories with a natural system, the Baues–Wirsching linear extension $D \rightarrowtail E_\omega \twoheadrightarrow S$.

D1 §2 l.243–246: "$[\omega] \in H^2(S; D)$, a Baues–Wirsching 2-cocycle ...; the chain splits as the claimed product exactly when $[\omega] = 0$ [12]." **Read**, because [12] is unseen. Your email states what [12] proves: $K = C \bowtie D \iff (L) \wedge (G)$ and $(G) \iff [\omega] = 0$. By that statement, $[\omega] = 0$ gives only (G), so "splits exactly when $[\omega] = 0$" is false in the direction $[\omega] = 0 \Rightarrow$ split unless (L) is a standing hypothesis. The note never states (L). **Gap (A1b).**

D2 §2 l.247–249: "Its defining cocycle condition is the associativity equation of the assembled composition, a system on N_3." **Checked-sober, with a hypothesis.** Write $p = (x, u)$, $q = (y, v)$, $r = (z, w)$. Then $(pq)r$ and $p(qr)$ have first coordinates $(xy)z$ and $x(yz)$, and second coordinates that differ by exactly $(d\omega)(x, y, z)$. That comparison needs the base composition to be associative and the action to satisfy $(xy){\cdot}w = x{\cdot}(y{\cdot}w)$. So: *over a category* S, $d\omega = 0 \iff E_\omega$ is associative. Both directions hold. **Computed** for every normalized 2-cochain in four cases (§3, T1).

D3 §2 l.249–251: "a cochain that violates it is not a weld at all, and there is then no class to form." **Checked-sober.** This is correct, and it is the N2 fix.

D4 §2 l.251–252 and Rem. 13 l.442–444: "what it classifies, by Eilenberg–Mac Lane, is whether the associative weld splits into its matched-pair factorization $C \bowtie D$ — the global closure condition (G), with $[\omega] = 0 \Leftrightarrow (G)$." Take the standard part first. For cocycles, $[\omega] = 0 \iff E_\omega$ admits a functorial section $\iff E_\omega \cong D \rtimes S$.

- ($\Leftarrow$) The cocycle of a functorial section is 0.
- ($\Rightarrow$) If $\omega = d\beta$, then $s(x) = (x, -\beta(x))$ is functorial.

Both directions use only D2's hypotheses. **Checked-sober** by hand, and **computed** (§3, T2), including a nontrivial action: Q_8 is non-split and D_8 is split.

Then the non-standard part.

- The phrase "splits into its matched-pair factorization $C \bowtie D$" identifies *split extension* with *Zappa–Szép product*. A split extension $D \rtimes S$ is a Zappa–Szép product in which one factor is normal and one action is trivial. A general matched pair is not an extension at all. **Computed**: $S_4 = S_3{\cdot}C_4$ is an exact factorization, and neither factor is normal. So "by Eilenberg–Mac Lane" covers only the semidirect case. The rest is [12]'s theorem, not EM's. (A-S1.)
- "splits" is equated with (G), but your own [12] says splitting is $(L) \wedge (G)$. (A1b.)

D5 §2 l.240–242: "Whether a candidate matched pair of actions closes into such a factorization is governed by the weld obstruction $[\omega]$." **This is the step that uses associativity it has not established.** D3 says the class exists only after the weld is known to be associative. So $[\omega]$ cannot decide whether an arbitrary *candidate* closes. It decides only whether an already-associative candidate splits. The note never says what makes the candidate weld associative, i.e. what makes ω a cocycle. Is it the matched-pair axioms? Is it (L)? Is it a theorem of [12]? **Computed**: over

$\mathbb{Z}/3$ with $\mathbb{Z}/2$ coefficients, 12 of the 16 normalized 2-cochains are *not* cocycles, and each gives a non-associative weld with no class. So the hypothesis has content. (A1c.)

D6 Rem. 13 l.440–441: "The 2-cocycle condition $d^2\omega = 0$ is the associativity equation of Theorem 12." **Wrong equation.** Theorem 12's equation is associativity of the *base* composition, a 3-element magma. The cocycle condition is associativity of the *twisted* composition over a base that is already a category (D2). Over Theorem 12's magma there is no cochain complex to speak of. **Computed**: $d \circ d \neq 0$ there, e.g. $\beta = (0, 0, 1)$ with $\mathbb{Z}/3$ coefficients gives $d(d\beta)(x, x, x) = 1$. So there is no H^2 for a class to live in. Conversely, the magma is not a cocycle failure of any twisted composition: 3 is prime, and every order-3 twisted composition is $\mathbb{Z}/3$ (**computed**, T4). The negative control breaks D2 when the base is not associative: over the magma, $\omega = 0$ trivially satisfies "$d\omega = 0$", yet the weld is non-associative (**computed**). (A1a.)

D7 Rem. 13 l.444–446 and §2 l.252–254: "$[\omega] = \varepsilon \neq 0$ ... is accordingly a non-split but associative weld: the cocycle condition is met, so associativity holds." **Checked-sober.** Both directions used are the valid ones: cocycle $\Rightarrow$ associative (D2), and split $\Rightarrow [\omega] = 0$ (D4 $\Leftarrow$), contraposed. This assumes [13]'s Lean certificate checks $d\omega = 0$ and does not only evaluate ω. Please confirm in one clause. **Computed** model: $\omega(1, 1) = 1$ on $\mathbb{Z}/2$ gives $\mathbb{Z}/4$, which is associative, non-split, and not an internal product of two order-2 subgroups.

Summary of the derivation. The skeleton is right and every implication points the right way:

$$\text{base a category} \;\Rightarrow\; \big(d\omega = 0 \iff E_\omega \text{ associative}\big), \qquad d\omega = 0 \;\Rightarrow\; \big([\omega] = 0 \iff E_\omega \text{ split}\big).$$

What the text still gets wrong is the *interface* with your weld. It names the wrong associativity equation (D6). It does not say where the weld's associativity comes from (D5). It drops (L) from "$\iff$ splits" (D1, D4). And it does not say which category is E_ω and which is the base (A2).

3 Computations (independent code, from the printed definitions)

All three scripts and their logs are in `scripts/2026-10-11-nerve-r2/`. Every negative control is built to flip a verdict, not to relabel an input.

Script	What it establishes (**computed**)
`check_magma_and_splitting.py`	Thm 12 table: total, unit 1, non-associative. Failures (x, x, x) and (x, y, x). 81 unital tables on $\{1, x, y\}$, of which 11 are associative, so the N_2/N_3 separation is strict. **Neg. control (breaks "non-associative")**: 3 one-entry mutations of the table are associative, and the checker reports them so. T1/T2, exhaustive over normalized cochains: $S = \mathbb{Z}/2$, $\mathbb{Z}/3$, Klein with $\mathbb{Z}/2$ coefficients, and $\mathbb{Z}/2$ acting on $\mathbb{Z}/4$ by sign. In every case cocycle $\iff$ associative, and for cocycles coboundary $\iff$ split. $\lvert H^2\rvert = 2, 1, 8, 2$. **Neg. control (breaks T1)**: over the magma, $\omega = 0$ yet the weld is non-associative. T3: $d \circ d \neq 0$ over the magma, and $= 0$ over $\mathbb{Z}/3$. T4: the magma is not a twisted composition. T5: $S_4 = S_3 \cdot C_4$ with neither factor normal.

Script	What it establishes (**computed**)
`check_N1_and_ levels.py`	Categories built from objects, arrows, identities and composition, axioms checked. $U([1]) = U(B(\mathbb{Z}/2) \sqcup \mathrm{pt}) = y^2 + y$. Their reflexive graphs are not isomorphic, and H^0 gives 1 vs 2 components. Brute force: exactly 3 categories have $U = y^2 + y$, in 2 tr_1-classes. **Neg. control (breaks "U-equal $\Rightarrow$ tr_1 separates")**: $\mathbb{Z}/2$ vs $\{1, e\}$ have equal U and *isomorphic* tr_1, and tr_2 separates them. Rem. 8: $H^n(-;\mathbb{Z})$ for $n \leq 3$ via Smith normal form gives first separation at degree 2. Over $\mathbb{F}_2$ the first separation is at degree 1.
`check_noteB_ rank1_lift.py`	For $\delta(\varepsilon) = k\,\varepsilon \otimes \varepsilon$ computed in $W^{\otimes r}$: $c\ell = \ell$ and coassociativity hold for $k = 0..3$. Universality $v(a; u, w) = (a, u, kw)$ is bijective iff $k = 1$. **Neg. control**: $k = 0, 2$ fail universality. Footnote 1: no bijection $K \to K \times K$ for $2 \leq \lvert K \rvert \leq 4$. Cantor: $\beta_1\beta_1 \neq \beta_1$ first at $k = 3$.

4 Note B: is the revision wording only?

I diffed the sentence-normalized `pdftotext` of `75603ad` against `6089268`. Every hunk falls into one of these groups:

- page-number and line-break artefacts;
- the p. 1 header (revision line, commit, recipient);
- the abstract and §5 "abstractly solid" wording;
- Prop. 12;
- step (4) "c-twist term is idle";
- footnote 1 (replaced);
- Rem. 15;
- Thm 18's opening sentence (added cite);
- the §8 and §9 scoping.

Theorem 1, Definition 4, Lemmas 7, 9, 10, 16 and 17, Theorems 11, 13, 18 and 22, and Remarks 14, 19, 20, 21 and 23 are textually unchanged up to layout. **No statement changed silently.** The one statement that did change is **Proposition 12**: its title is now "abstractly solid", and it drops "hence an isomorphism ℓ_M" and "solidity, read as invertibility of ℓ_M, does not by itself bound the rank". This is the change we asked for (old M3). It weakens a false clause to a true one and leaves every graded result untouched. The p. 1 parenthesis "no statement changed" is therefore inexact. Say "Prop. 12 restated as requested". The registry grades from our 75603ad report stand: Thm 22 module form proved, Thms 11/18 proved, "$\mathbb{N} \cdot y$ carries no tangent structure" proved.

5 Items and verdicts

Note A

A1 MUST-FIX Splitting derivation: three repairs (§2 l.240–254, Rem. 13 l.438–448).

(a) Rem. 13: replace "The 2-cocycle condition $d^2\omega = 0$ is the associativity equation of Theorem 12" with: "the 2-cocycle condition $d\omega = 0$ is associativity of the *twisted* composition over the already-associative base S. Theorem 12 is a failure of associativity of the base itself, over which no cochain complex exists ($d \circ d \neq 0$). Both are N_3-predicates, on different objects."

(b) §2 l.246 and Rem. 13 l.443–444: by your own [12], $[\omega] = 0 \iff (G)$ and splitting $\iff (L) \wedge (G)$. Either state (L) as a standing hypothesis ("given (L), the chain splits iff $[\omega] = 0$"), or replace "splits" with "satisfies (G)".

(c) §2 l.240–242: say what makes the candidate weld associative, i.e. what makes ω a cocycle: the matched-pair axioms, (L), or a lemma of [12]. Then change "whether a candidate matched pair closes ... is governed by $[\omega]$" to "whether an associative candidate splits". As written, the class is said to decide something it presupposes.

A2 MUST-FIX N2′, carried over and now sharper. Under the new gloss, "$[\omega] = 0 \iff S$ splits as $C \bowtie D$". Theorem 14 assumes $S = C \bowtie D$, so within that hypothesis $[\omega] = 0$ always. Yet Theorem 15 and Example 16 need $[\omega] \neq 0$. The EM reading resolves this, and you should say it in one line: $[\omega] \in H^2(S; D)$ is a class *of the base* S. It classifies an extension $D \rightarrowtail E_\omega \twoheadrightarrow S$, and it is E_ω that splits or not. For the rigid-twist witness, $S = B(\mathbb{Z}/2)$, $D = \mathbb{Z}/2$ and $E_\omega = B(\mathbb{Z}/4)$. Also resolve the clash between D as a wide subcategory (l.239) and D as the vertex-group coefficient system (l.243–245).

A3 MUST-FIX N3 residue: Remark 8 contradicts your new definition. p. 4 l.180–181 defines the nerve level of an H^n class as $n+1$. The integral detector $H^2(\mathbb{Z}/2; \mathbb{Z})$ therefore has nerve level 3, and the $\mathbb{F}_2$ detector has nerve level 2 (**computed**). So l.363–364, "which nerve level always does [have a coefficient-free answer]: N_2", is false as defined, and so is the §7 reliance (l.542, "nerve level is coefficient-free"). The coefficient-free object is the level of the *datum* $a \cdot a = 1$, a raw datum on N_2. Say: "the datum sits at nerve level 2 for every coefficient ring; the detecting *class* sits at nerve level 2 over $\mathbb{F}_2$ and 3 over $\mathbb{Z}$". Fix W1 in the same edit.

A4 SHOULD Cite Baues–Wirsching (linear extensions are classified by $H^2(C; D)$) rather than "by Eilenberg–Mac Lane". Say that EM-split means semidirect, and that the general matched-pair statement is [12]'s (D4).

A5 SHOULD Carry over W2 (abstract l.24–26, §7 l.514–516, l.573–575), W5 (l.376–377), and C, D for the diamond in Example 16.

A6 OK N1 (**computed**). Recipient. Cochain level vs nerve level in Remark 2. Theorem 12's magma (**computed**). The re-entrancy direction (D7, **checked-sober**, modulo [13] checking $d\omega = 0$). Cosmetic: "$d^2\omega$" reads as $d \circ d$, so write $d\omega$ or $d^2(\omega)$ with the degree as a subscript. Remark 2's "the weld $[\omega] \in H^2$ is a 2-cochain": a class is not a cochain, so say "is represented by". W3 and W4 are optional.

Note B

B1 MUST-FIX Old M3 residue in §1 (p. 4 l.172–185). "It is true that $\mathbb{N} \cdot y$ is solid: ... supplies an isomorphism $M \otimes M \cong M$, hence an isomorphism ℓ_M" contradicts Def. 4 (l.278: solid means ℓ_M

is an isomorphism) and Theorem 13. Change it to "abstractly solid ... an abstract isomorphism, not a vertical lift". Add "abstract" at l.167, l.183 and l.202.

B2 MUST-FIX Old M1 residue in §1 (p. 2 l.75–76). Change "This note proves there cannot" to "... there cannot, among representable (Weil base-change) structures on the linear layer".

B3 SHOULD S1 to S5 as in our 75603ad report. They are unchanged except S1's "idle" wording and the misplaced S3 cite. S3 is now **computed**: the closing of Theorem 18 and Rem. 23 must cite universality (Theorem 13 step (1)), because $c\ell = \ell$ and coassociativity do not pin k.

B4 OK M2 (footnote), M4 (recipient), Proposition 12 restatement, and the §8/§9 scoping. **No silent statement change.** Cosmetic: C1–C7 and the nits are still open, and the p. 1 "no statement changed" should mention Prop. 12.

Weld nodes: recommendation

Keep both weld nodes demoted on 6089268. After A1 and A2 land, the gloss "cocycle condition = associativity of the twisted weld; $[\omega]$ = splitting obstruction" earns **checked-sober**. Its standard core is **computed** here in both directions. The identification $(G) \iff [\omega] = 0$ stays **read** until [12] is attached. I would not grade any weld node *proved* on the strength of this note, which is a positioning note citing [12] and [13].